

Spencer R. Keyser

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Education

Ph.D., Wildlife Ecology **2020 – 2024**

Department of Forest and Wildlife Ecology
 University of Wisconsin-Madison – Madison, WI

M.S., Marine Science **2017 – 2019**

Marine Science Institute
 University of Texas at Austin – Austin, TX

B.S., Wildlife Ecology **2013 – 2017**

Certificate in Environmental Studies

Department of Forest and Wildlife Ecology
 University of Wisconsin-Madison – Madison, WI

Professional Appointments

Research Associate **2026 – Present**

Center for Engagement in Science and Nature
 Cornell Lab of Ornithology
 Cornell University – Ithaca, NY

Postdoctoral Researcher **2024 – 2026**

Cornell K. Lisa Yang Center for Conservation Bioacoustics
 Cornell Lab of Ornithology
 Cornell University – Ithaca, NY

In-Review/In-Prep Publications

Wood C. M., **Keyser S. R.**, Beatty M. A., and K. Stewart. Passive acoustic monitoring for estimating Essential Biodiversity Variables for ecosystem management. In-prep

Keyser S. R., Smith M., Notaro M., Zuckerberg B., and J. N. Pauli. Warming winters threaten the rewilding and recovery of North America's most translocated carnivore. In-review: *Global Change Biology*

Perea S., **Keyser S. R.**, Buron R., Berman L., Boucher T., Koto S. E., Froese G., Gottesman B. L., Ebanega M. E., Mbamy W., Yoh T., Fandos G., and Z. Burivalova. Management-unit context outweighs certification status in shaping Central African bird communities. In-Prep

Keyser S. R., Peery M. Z., Chaon P., Kramer A., and C. M. Wood. Fire has not yet restructured avian assemblages across the Sierra Nevada. In-review: *Frontiers in Ecology and the Environment*.

Peer-reviewed Publications

Keyser S. R., Pauli J. N., Fink D., Radeloff V. C., and B. Zuckerberg. Hotspots in cold climates: Mapping and evaluating the conservation of cold-adapted biodiversity across the United States. 2026. *Diversity and Distributions*.

*Bielski L., **Keyser S. R.**, Peery M. Z., and C. M. Wood. Warm temperatures and severe fire drive a range contraction of an avian old-forest specialist in the Sierra Nevada, USA. 2026. *Frontiers in Bird Science*. <https://doi.org/10.3389/fbirs.2026.1737313>

Keyser S. R., Pauli J. N., Fink D., Radeloff V. C., Pigot A. L., and B. Zuckerberg. Seasonality structures avian functional diversity and niche packing across North America. 2024. *Ecology Letters*. <https://doi.org/10.1111/ele.14521>

Keyser S. R., Fink D., Gudex-Cross D., Radeloff V. C., Pauli J. N., and B. Zuckerberg. Snow cover dynamics: An overlooked yet important feature of overwinter bird occurrence and abundance across the United States. 2022. *Ecography*. <https://doi.org/10.1111/ecog.06378>

Gilbert N. A., McGinn K., Nunes L., Shipley A., Bernath-Plaisted J., Clare J., Murphy P., **Keyser S. R.**, Thompson K., Maresh Nelson S., Cohen J., Widick I., Bartel S., Orrock J., and B. Zuckerberg. Diel activity patterns as an emerging means to explore ecological processes. 2022. *Trends in Ecology and Evolution*. <https://doi.org/10.1016/j.tree.2022.10.008>

Oke T., Zhang S., **Keyser S. R.**, and L. A. Yeager. Sea-surface temperature anomalies mediate changes in fish richness and abundance in Atlantic and Gulf of Mexico estuaries. 2022. *Journal of Biogeography*. <https://doi.org/10.1111/jbi.14451>

Gudex-Cross D., Zhu L., **Keyser S. R.**, Zuckerberg B., Pauli J. N., and V. C. Radeloff. Winter conditions structure extratropical patterns of species richness of amphibians, birds, and mammals globally. 2022. *Global Ecology and Biogeography*. <https://doi.org/10.1111/geb.13511>

Gudex-Cross D., **Keyser S. R.**, Zuckerberg B., Fink D., Pauli J. N., and V. Radeloff. Winter Habitat Indices (WHIs) for the contiguous US and their relationship with winter bird diversity. 2021. *Remote Sensing of Environment*, 255, 112309. <https://doi.org/10.1016/j.rse.2021.112309>

Yeager L. A., Estrada J., Holt K., **Keyser S. R.**, and T. Oke. Are habitat fragmentation effects stronger in marine systems? A review and meta-analysis. 2020. *Current Landscape Ecology Reports*, 5, 58-67. <https://doi.org/10.1007/s40823-020-00053-w>

Biggs C. R., Yeager L. A. Bolser D. G., Bonsell, C., Dichiera A. M., Hou Z., **Keyser S. R.**, Khursigara A. J., Lu K., Muth A., Negrete B., and B. Erisman. Does functional

redundancy affect ecological stability and resilience? A meta-analysis. 2020. *Ecosphere*, 11(7), e03184. <https://doi.org/10.1002/ecs2.3184>

*Student-mentee led paper

Contributed Presentations (Oral)

Keyser S. R., Peery M. Z., Kramer A., Winiarski J., and C. M. Wood. "Investigating the drivers of avian beta-diversity across the Sierra Nevada, CA." American Ornithological Society Annual Meeting. St. Louis, MO. August 2025.

***Keyser S. R.**, Pauli J. N., Fink D., Radeloff V. C., Pigot A. L., and B. Zuckerberg. "Seasonal patterns and drivers of North American functional diversity." American Ornithological Society-Society for Canadian Ornithologists Joint Conference. London, Ontario. August 2023.

Zuckerberg B., **Keyser S. R.**, Gudex-Cross D., Radeloff V. C., Fink D. and J. N. Pauli. "Applying Sustainable Development Goals to the conservation of winter environments and cold-adapted species in a warming world." NASA Biodiversity Ecological Forecasting Meeting. College Park, MD. September 2022.

Gudex-Cross D., **Keyser S. R.**, Zuckerberg B., Fink D., and V. C. Radeloff. "Winter Habitat Indices (WHIs) for the contiguous US and their relationship with winter bird biodiversity." Ecological Society of America annual meeting. August 2020.

Keyser S. R., J. Cruz, C. Giri, and L. A. Yeager. "Impacts of regional climate change and climate-mediated vegetation shifts on coastal avian community dynamics." Ecological Society of America annual meeting. Louisville, KY. August 2019.

Yeager L. A., Estrada J., Holt J., **Keyser, S. R.**, and O. Oke. "Are effects of habitat fragmentation stronger in marine systems? A literature review and meta-analysis." Ecological Society of America annual meeting. Louisville, KY. August 2019.

Keyser S. R. and L. A. Yeager. "Impacts of climate change on coastal avian biodiversity." Graduate Climate Conference. Seattle, Washington. November 2018.

Keyser S. R. and J. Chaves. "Neighbor-Stranger recognition and song alteration in White-eared Ground - sparrows (*Melospiza leucotis*)." CIEE Tropical Ecology and Conservation Spring 2016 Research Symposium. Monteverde, Costa Rica. May 2016.

Allstadt A., Caraco T., Korniss G., and **S. R. Keyser**. "Kinetic roughening theory predicts properties of ecological fronts in a variety of systems and spatial scales." Ecological Society of America annual meeting. Baltimore, MD. August 2015.

*Presentation selected for award

Contributed Presentations (Poster)

Keyser, S. R., Fink, D., Gudex-Cross, D., Radeloff, V. C., Pauli, J. N., and B. Zuckerberg. "Assessing conservation gaps for U.S. winter biodiversity." American Geophysical Union. Chicago, IL. December 2022.

Keyser, S. R., Fink, D., Gudex-Cross, D., Radeloff, V. C., Pauli, J. N., and B. Zuckerberg. "Snow cover dynamics as a key ecological filter for species distributions." NASA Biodiversity & Ecological Forecasting Meeting. College Park, MD. September 2022.

Keyser, S. R., Fink, D., Gudex-Cross, D., Radeloff, V. C., Pauli, J. N., and B. Zuckerberg. "Snow cover dynamics as a key ecological filter for species distributions." Midwest Climate Adaptation Science Center Annual Meeting. Madison, WI. August 2022.

Keyser, S. R., Fink, D., Gudex-Cross, D., Radeloff, V. C., Pauli, J. N., and B. Zuckerberg. "Snow cover dynamics as a key ecological filter for species distributions." American Geophysical Union. New Orleans, LA. December 2021.

Keyser, S. R. and Yeager, L. A. "Impacts of climate-mediated mangrove expansion on avian community dynamics." International Ornithological Congress. Vancouver, British Columbia. August 2018.

Keyser, S. R. and Yeager, L. A. "Impacts of climate-mediated mangrove expansion on avian community dynamics." Benthic Ecology Meeting. Corpus Christi, TX. March 2018.

Invited Public Talks

Keyser, S. R. "Winter warriors: How birds survive the elements." Lake/Cook Audubon Society. Virtual Seminar. January 2021.

Keyser, S. R., Cruz, J., Giri, C., and L. A., Yeager. "Impacts of climate change on bird biodiversity across the Gulf of Mexico." Coastal Bend Audubon Society. Corpus Christi, TX. November 2019.

Fellowships & Grants

NASA FINESST Fellowship (\$100,000)	2023
GCA Francis M Peacock Scholarship for Native Bird Habitat (\$5,000)	2019
Coastal Bend Audubon Society Bird Conservation Research Award (\$500)	2018

Awards

AOS Best Student Presentation Award	2023
University of Texas Marine Science Institute Graduate Travel Award	2019
Graduate Climate Conference Presenter Travel Award	2018

Professional Training

Structural Equation Modeling: Advanced Techniques	2018
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Ecological Society of America Annual Meeting Workshop
Louisville, KY

Introduction to Applied Bayesian Statistics 2018
Department of Statistics and Data Sciences
University of Texas at Austin, Austin, TX

Applied Hierarchical Linear Modeling 2018
Department of Statistics and Data Sciences,
University of Texas at Austin, Austin, TX

Teaching & Mentoring

Advanced Bioacoustics Workshop Co-lead 2026
Indian Institutes of Science and Research (IISER) Tirupati
Tirupati, Andhra Pradesh, India
Advanced Bioacoustics Workshop

Teaching Assistant 2021-2023
Department of Forest and Wildlife Ecology
University of Wisconsin-Madison
INTEGSCI 110: BioHouse Seminar: Biology for the 21st Century

Teaching Assistant 2019
Department of Marine Science
University of Texas at Austin
MNS 120L: Field Studies in Marine Ecology

Teaching Assistant 2018
Department of Marine Science
University of Texas at Austin
MNS 307: Introduction to Oceanography

Professional Experience

Wildlife Field Technician 2017
Kentucky Department of Fish and Wildlife

Undergraduate Research Technician 2016-2017
SILVIS Lab
University of Wisconsin-Madison

Spotted Owl Field Technician 2016
Peery Lab
University of Wisconsin-Madison

Professional Affiliations

American Ornithological Society 2023-present

American Geophysical Union	2020-2023
Ecological Society of America	2017-2020
International Ornithologists' Union	2017-2019
The Waterbird Society	2017-2019
The Wildlife Society	2015-2017

Professional Service

Scientific Journal Service

Reviewer for *Biological Conservation*
Reviewer for *Journal of Mammalogy*
Reviewer for *Biological Conservation*
Reviewer for *Global Ecology and Biogeography*
Reviewer for *Conservation Biology*
Reviewer for *Oikos*
Reviewer for *Oecologia*
Reviewer for *Proceedings to the Royal Society B*

University Service

Graduate Representative
Dept. Forest and Wildlife Ecology Justice, Diversity,
Equity and Inclusion (JEDI) Committee

2022-2024

Board Member
UT Marine Science Graduate Student Association

2018-2019

Community Outreach

Summer Science Instructor
University of Texas Marine Science Institute

2019

Science Fair Judge
Olsen Elementary School, Port Aransas, TX

2018